

RD INFO TECHNOLOGY – DATA SCIENCE INTERNSHIP

TASK 1

BUSINESS PROBLEM UNDERSTANDING

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Project Domain

Data Science & Machine Learning



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Project Description

This project focuses on understanding customer churn problems in online businesses and converting business requirements into a machine learning problem using data science concepts and business analytics techniques.

Introduction

Business Problem Understanding is the first and one of the most important stages in every data science project. Before building machine learning models or analyzing data, it is necessary to clearly understand the actual business problem, company goals, customer needs, and expected outcomes.

In this task, the focus is on identifying a real-world business issue and converting it into a data science problem that can be solved using data analysis and machine learning techniques.

This task helps in understanding how businesses use data science to improve decision-making, increase profits, reduce losses, and improve customer satisfaction.

Business Problem

Nowadays, many online shopping and e-commerce companies face customer churn problems. Customer churn means customers stop using the platform, cancel subscriptions, or move to competitor platforms.

This creates major business challenges because losing customers directly affects:

- Company revenue
- Customer engagement
- Brand growth
- Profitability
- Market reputation

The company wants to identify customers who are likely to leave the platform in advance so that proper actions can be taken to retain them.

Some common reasons for customer churn are:

- Poor customer service
- High prices
- Lack of engagement

- Delayed delivery
- Better competitor offers
- Bad user experience

If businesses can predict customer churn early, they can:

- Provide personalized offers
- Improve services
- Increase customer satisfaction
- Reduce customer loss
- Improve business performance

Therefore, understanding customer behavior is an important business requirement.

Objective of the Project

The main objective of this task is to understand the business problem clearly and convert it into a structured data science problem.

The objectives include:

1. Understanding the company's business goals.
2. Identifying the major business issue.
3. Defining project requirements.
4. Understanding customer-related challenges.
5. Identifying important KPIs (Key Performance Indicators).
6. Converting the business problem into a machine learning problem.
7. Understanding how data can help in decision-making.

The overall goal is to help the organization make better business decisions using data science techniques.

Understanding the Business Requirements

Before starting any data science project, it is important to communicate with stakeholders, managers, and business teams to understand the exact requirements.

The following questions are usually discussed:

- What problem is the company facing?
- What are the business goals?
- What type of customers are affected?
- What data is available?
- What outcomes are expected?
- How will the solution help the company?

Understanding these requirements helps in building accurate and useful solutions.

Key Performance Indicators (KPIs)

KPIs are measurable values used to evaluate business performance.

The important KPIs for this project are:

1. Customer Retention Rate

Measures how many customers continue using the platform over time.

Formula:

Customer Retention Rate =
 $(\text{Number of Customers Retained} / \text{Total Customers}) \times 100$

2. Customer Satisfaction Score

Measures customer happiness and service quality.

Higher satisfaction generally leads to lower churn.

3. Monthly Sales Growth

Measures increase or decrease in company sales every month.

This helps understand business performance trends.

4. Active User Count

Shows how many users actively use the platform regularly.

Low activity may indicate possible customer churn.

5. Customer Engagement Rate

Measures customer interaction with the platform through purchases, visits, reviews, or activity.

Higher engagement means stronger customer relationships.

Converting Business Problem into Data Science Problem

Business problems are converted into data science problems so they can be solved using data analysis and machine learning.

Business Problem:

Customers are leaving the online shopping platform.

Data Science Problem:

Build a machine learning model that predicts whether a customer is likely to leave the platform based on customer behavior data.

This becomes a classification problem in machine learning.

The system analyzes:

- Purchase history
- Login activity
- Customer reviews
- Browsing behavior
- Transaction frequency
- Customer support interactions

Using this data, the model predicts churn probability.

Data Required for the Project

The following types of data may be required:

Data Type	Description
Customer ID	Unique customer identification
Purchase History	Customer buying details
Login Activity	User platform usage frequency
Customer Reviews	Feedback and ratings
Transaction Records	Payment and order history
Support Tickets	Customer complaints/issues
Subscription Status	Active or inactive membership

Tools Used

The following tools are used during business problem understanding:

1. Meetings

Used to communicate with stakeholders and business teams.

2. JIRA

Used for project management and task tracking.

3. MS Word

Used for documentation and reporting.

4. MS Excel

Used for organizing and analyzing business data.

Expected Outcome

The expected outcome of this task is:

- Better understanding of the business issue
- Identification of customer churn patterns
- Clear project objectives
- Well-defined KPIs
- Conversion of business problem into machine learning problem
- Improved business decision-making process

The final solution can help companies:

- Reduce customer churn
- Improve customer retention
- Increase profits
- Improve customer experience

Learning Outcome

Through this task, the following concepts were learned:

- Understanding real-world business problems
- Importance of KPIs
- Business requirement analysis
- Stakeholder communication
- Data science project planning
- Problem statement creation and Conversion of business problems into ML problems

Conclusion

Business Problem Understanding is the foundation of every successful data science project.

Without clearly understanding the business goals and requirements, it becomes difficult to build effective machine learning solutions.

In this task, the business issue related to customer churn was analyzed, KPIs were identified, and the business challenge was successfully converted into a data science problem.

This task provided practical knowledge about how companies use data science for solving real-world business challenges and improving decision-making processes.

Thank You

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